7.H1.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on the Law of Sines?

Unit 7, Lesson H2: Law of Cosines

Benchmark

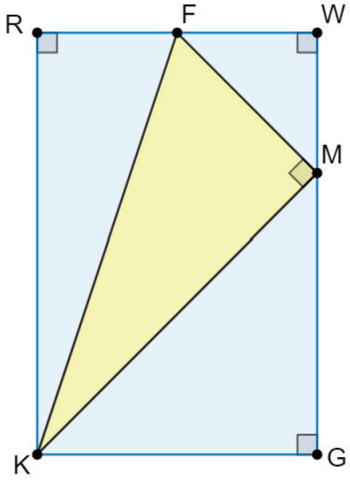
MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

Learning Target

- [ ] I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.

7.H2.1 Warm-Up: Reasoning with Triangles

1. A diagram of a triangle inscribed in a rectangle is shown.

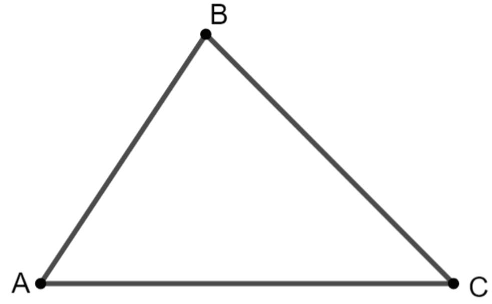


   a. Liu Yang told her partner that all of the right triangles in the diagram are similar. Determine if the diagram contains enough information to verify Liu Yang's statement. Explain your reasoning.

   b. In the diagram, $m\angle RFK = 72°$ and $m\angle MKF = 63°$. Use these measures to evaluate your reasoning. If necessary, make adjustments.

7.H2.2 Collaboration: Deriving the Law of Cosines

1. A triangle is shown below.



   a. Use ΔABC to complete the following.

   - Sketch an altitude from vertex C.
   - Label the altitude h, and label the point on $\overline{AB}$ as D.
   - Label each line segment. $AC = b,\ AD = x, CB = a,$ $DB = c - x,\ AB = c$

The altitude creates two right triangles inside ΔABC. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

   b. Use the labels on the two right triangles inside ΔACB to complete the table.

ΔADC	ΔCDB
$\_\_\_\_^2 + \_\_\_\_^2 = \_\_\_\_^2$	$\_\_\_\_^2 + \_\_\_\_^2 = \_\_\_\_^2$

c. What do you notice about both equations?

d. Rearrange the Pythagorean Theorem for ΔADC in terms of h^2.

e. Substitute the new equation in place of h^2 in ΔCDB.

f. Use cosine to write the trigonometric ratio for $\angle A$.

g. Solve the trigonometric ratio for $\angle A$ for x.

h. Substitute $b \cos A$ in place of x in the equation from part e.

i. Expand and combine like terms.

7.H2.3 Guided Instruction: Law of Cosines

The **Law of Cosines** can be written as one of the following:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

1. In ΔDRP, $DR = 10,\ DP = 4.7,$ and $m\angle RDP = 60°$.

   a. Draw and label a triangle with the given measurements.

   b. Use the law of cosines to find the measure of $\overline{RP}$.

   c. Use the law of cosines to find $m\angle RPD$.

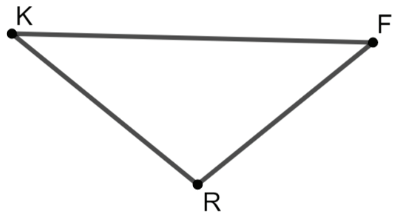
2. The measurements of the sides of ΔRLP are $RL = 4.5$, $PL = 7.2$, and $RP = 6.7$. Determine the measures of the angles of ΔRLP.

Angle	Measurement
$\angle L$	
$\angle P$	
$\angle R$	

7.H2.4 Guided Practice: Law of Cosines

1. ΔKFR is shown. $KF = 15$, $FR = 9$, and $KR = 10$. Find the missing angles.

Angle	Measurement
$\angle F$	
$\angle K$	
$\angle R$	



2. Two fishing boats begin traveling in the same direction from a port in Miami. One travels 20 nautical miles per hour on a bearing of $30°$ and the other travels 16 nautical miles per hour on a bearing of $45°$. Assuming each boat maintains its course and speed, how far apart will the fishing boats be after three hours?

7.H2.5 Your Turn!

1. In ΔBNW, $BN = 9$, $NW = 12$, and $BW = 13$. Find the missing angles.

Angle	Measurement
$\angle B$	
$\angle N$	
$\angle W$	

2. In ΔSLC, $SC = 34$, $SL = 22$, and $m\angle LSC = 42°$. Find the missing sides and angles.

Part	Measurement
$\overline{LC}$	
$\angle C$	
$\angle L$	

7.H2.6 Wrap-Up: Cool Down

1. Write a summary of what you learned today.

Unit 7, Lesson H3: Area of a Triangle Using the Sine Ratio

Benchmark

MA.912.T.1.4 Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.

Learning Target

- ☐ I can find the area of a triangle using the sine ratio.